INCH-POUND

A-A-59830
31 August 2009

COMMERCIAL ITEM DESCRIPTION

ANALYZER, DIGITAL COMMUNICATION

The General Services Administration has authorized the use of the commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a Digital Communications Analyzer. The Analyzer is a multifunction test instrument capable of performing a comprehensive set of measurements to evaluate the protocol-insensitive characteristics of digital communication systems. The instrument will be able to test data rates ranging from 50 bits per second (bps) to 15 megabits per second (Mbps) using a set of data interfaces including EIA-TIA232, RS-449, MIL-STD-188-114, EIA 530, V.35 ITU/306, T-Carrier, and Diphase/Conditioned Diphase. The analyzer will test both Data Terminal Equipment (DTE) and Data Circuit-Terminating Equipment (DCE) modes, when applicable. This CID is meant as a minimum requirement for the equipment in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet this CID. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The analyzer shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

2.2.1 Temperature Operating. The analyzer shall conform to its specified performance and accuracy while being operated in a temperature range of 0°C to +50°C.

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FSC 6625

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2.2.2 Temperature Non-operating. The analyzer shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -30°C to +75°C.

2.2.3 Humidity. The equipment shall meet its performance and accuracy requirements in an operating environment of 0 to 90% relative humidity.

2.2.4 Altitude Operating. The analyzer shall conform to its specified performance and accuracy while being operated in an altitude range of 0 to 10,000 feet.

2.2.5 Altitude Non-operating. The analyzer shall conform to its specified performance and accuracy after being stored in an altitude range of 0 to 10,000 feet.

2.3 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 115 volts AC (±10%) at line frequencies of 48 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 1500 hours of operation with a statistical certainty of 95%.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the analyzer.

2.6 Performance Requirements. The analyzer shall meet all requirements specified herein, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800 for the class specified herein.

2.6.1 Delay Measurement. The analyzer shall be capable of round trip delay measurement and clock data change measurement to support satellite testing

2.6.2 Clock Recovery. The analyzer shall be capable of analyzing any synchronous timing data interface measured by the analyzer, and extract clock information from the received data.

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2.6.3 Communication Interfaces Testing. The analyzer shall test the following digital communication interfaces: EIA-TIA232 DTE/DCE, RS-449 DTE/DCE, EIA-530 DTE/DCE, MIL-STD-188-114 DTE/DCE, V.35 ITU/306 DTE/DCE, T1/Fractional T1 with Drop and Insert, and Diphase/Conditioned Diphase.

2.6.3.1 EIA-TIA232 DTE/DCE. The analyzer shall comprehensively test digital communications DTE and DCE interfaces as defined by EIA-TIA232. At least one 25-pin connector shall be provided which can be programmed for DTE or DCE operation. Standard EIA-TIA232 connectors shall be supplied or adapters/cables shall be provided to allow standard EIA-TIA232 connections. Non-synchronous communications shall be tested at rates from 50 bits per second to 20,000 bits per second.

2.6.3.2 RS-449 DTE/DCE, EIA-530 DTE/DCE, and MIL-STD-188-114 DTE/DCE. The analyzer shall comprehensively test digital communications DTE and DCE interfaces as defined by RS-449, EIA-530, and MIL-STD-188-114. At least one each 25-pin and 37-pin connectors shall be provided which can be programmed for DTE or DCE operation. Non-synchronous communications shall be tested at rates from 50 bits per second to 20,000 bits per second.

Unbalanced synchronous communications shall be tested at rates from 50 bits per second to 100 kilobits per second. Unbalanced synchronous communications testing shall be capable of extracting clock information from the received data at rates from 50 bits per second to 100 kilobits per second.

Balanced synchronous communications shall be tested at rates from 50 bits per second to 15 megabits per second. Balanced synchronous communications testing shall be capable of extracting clock information from the received data at rates from 50 bits per second to 520 kilobits per second.

2.6.3.3 V.35 ITU/306. The analyzer shall comprehensively test digital communications DTE and DCE interfaces as defined by V.35 ITU/306. At least one 34-pin V.35 ITU/306 connector shall be provided which can be programmed for DTE or DCE operation. Standard V.35 ITU/306 connectors shall be provided or adapters/cables shall be provided to allow standard V.35 ITU/306 connections. Synchronous communications shall be tested at rates from 50 bits per second to 15 megabits per second.

2.6.3.4 T1/ Fractional T1. The analyzer shall comprehensively test the following T1 interfaces: Full T1, Fractional T1, Fractional T1 Drop and Insert, Voice, Test Loopback (TLB), Line Loopback (LLB), and ESF Data Link. The following connectors shall be provided: dual bantam for T1 xmit pair/rcv pair, bantam T1 reference input, ESF data link/DS0, and a handset connector. The analyzer shall test Full T1 at 1.544 Mbps unframed and at 1.536 Mbps framed. The analyzer shall test Fractional T1 at any Nx56 kilobits per second (kbps) or Nx64

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kbps data rate. Fractional Drop and Insert shall be performed on any Nx56 kbps or Nx64 kbps without disrupting traffic on remaining timeslots.

2.6.3.5 Diphase. The analyzer shall comprehensively test digital communications formatted as diphase or conditioned diphase, as defined in MIL-STD-188-200. The analyzer shall test diphase and conditioned diphase formatted signals for the following data rates: 1.2 kbps, 2.4 kbps, 4.8 kbps, 9.6 kbps, 16 kbps, 32 kbps, 64 kbps, 72 kbps, 128 kbps, 144 kbps, 256 kbps, 288 kbps, 512 kbps, 576 kbps, 1024 kbps, 1152 kbps, 1536 kbps, 2048 kbps, 2304 kbps, 4096 kbps, and 4608 kbps.

2.6.4 Expandability. The analyzer shall be capable of expansion to support other communications interfaces. Expansion shall not require internal hardware modification.

2.6.5 Remote Control. The analyzer shall have remote control capability. The remote control capability shall be a general serial interface for either EIA-TIA232 or USB2 or both. Remote control shall be possible independently through either interface.

2.6.6 Rack mount. The analyzer shall be capable of being rack-mounted in a standard 19 inch equipment rack with an optional rack-mount conversion kit.

2.6.7 Accessories. The following accessories, or equivalents, shall be included with the analyzer:

2.6.7.1 Cables.

2.6.7.1.1 EIA-TIA232/EIA-530, 25-pin D male-to-male, 10 feet in length.

2.6.7.1.2 V.35 ITU/306 34-pin male-to-male, 10 feet in length.

2.6.7.1.3 Crossover cable for V.35 ITU/306 DTE/DCE male-to-female, 9 inches in length.

2.6.7.1.4 RS-449/MIL-188 37-pin D male-to-male, 10 feet in length.

2.6.7.1.5 Bantam-to-Bantam, 10 feet in length - Quantity 2 .

2.6.7.2 Carrying Case. Soft carrying case.

2.6.8 Physical Characteristics. The analyzer shall be portable equipment with the following physical characteristics.

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2.6.8.1 Dimensions. The equipment shall be no larger than: 7.0 inches high, 19.0 inches wide, and 20 inches deep. The width shall be compatible with mounting in a standard 19 inch equipment rack.

2.6.8.2 Weight. The total weight of the equipment, including all accessories (excluding manuals) shall not exceed 25 lbs.

2.6.9 Manuals. The analyzer shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this PD, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this PD are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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4.3 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Department of Defense and Federal documents may be obtained at http://assist.daps.dla.mil or from the Document Automation and Production Service, Bldg 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia PA 19111-5094.

6.1.2 The Federal Acquisition Regulation (FAR) may be obtained at http://www.gpoaccess.gov/cfr/index.html or available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 EIA standards may be obtained at http://www.eia.org or available from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

6.1.4 ITU Telecommunication Standardization Sector (ITU-T). Documents can be obtained from: ITU, Place des Nations, CH-1211 Geneva 20, Switzerland. Telephone number: +41 22 730 51 11. Email: itumail@itu.int.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Product conformance provisions.
c. Packaging requirements.

6. 3 Key words.

Comprehensive measurements
Protocol-insensitive characteristics
Test instrument

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